

Examples of probability distributions in action

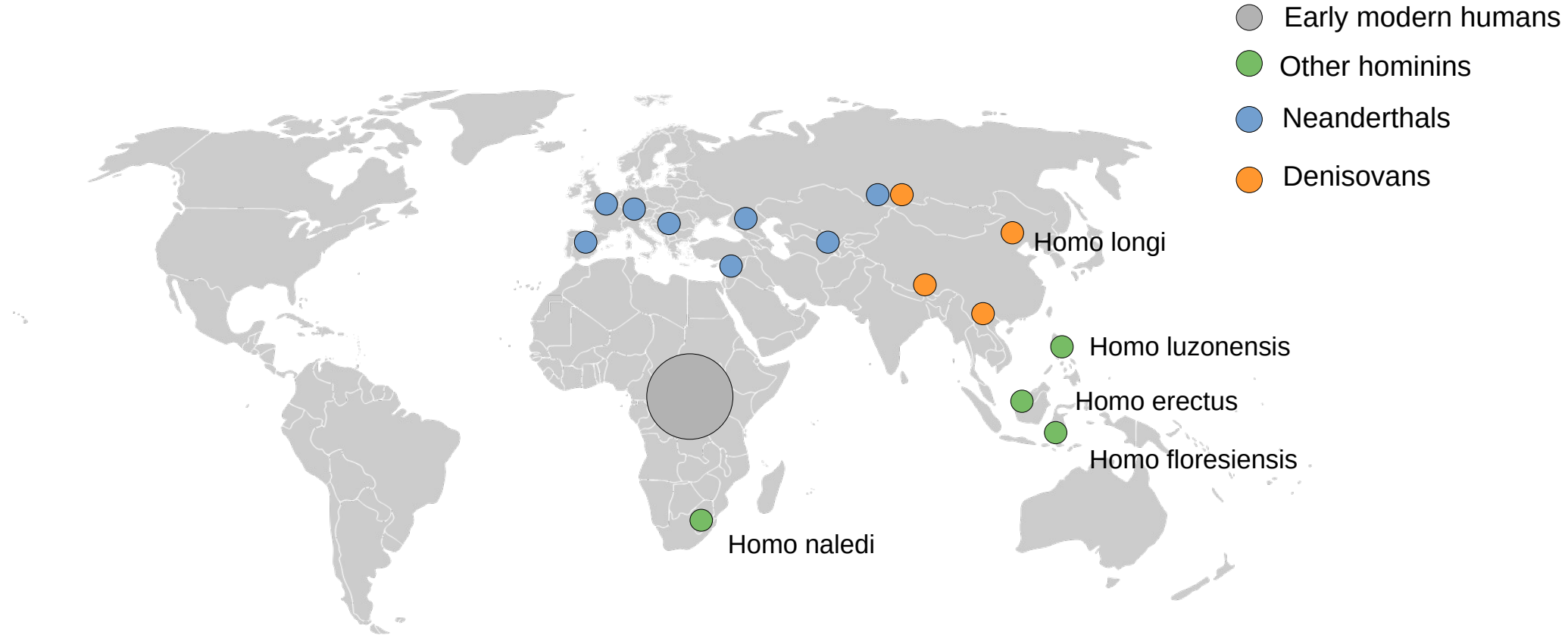
Reviving extinct archaic humans



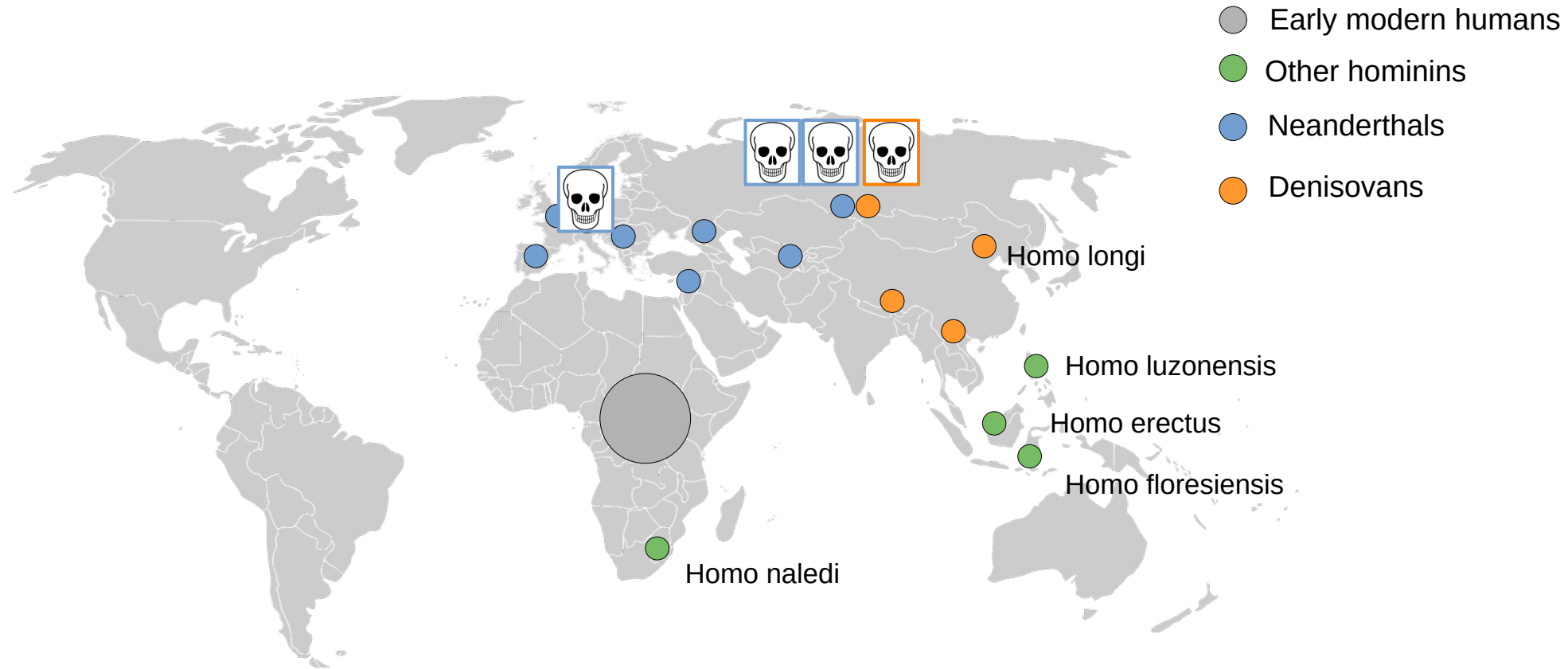
My research

Illustrations by Tom Bjorklund

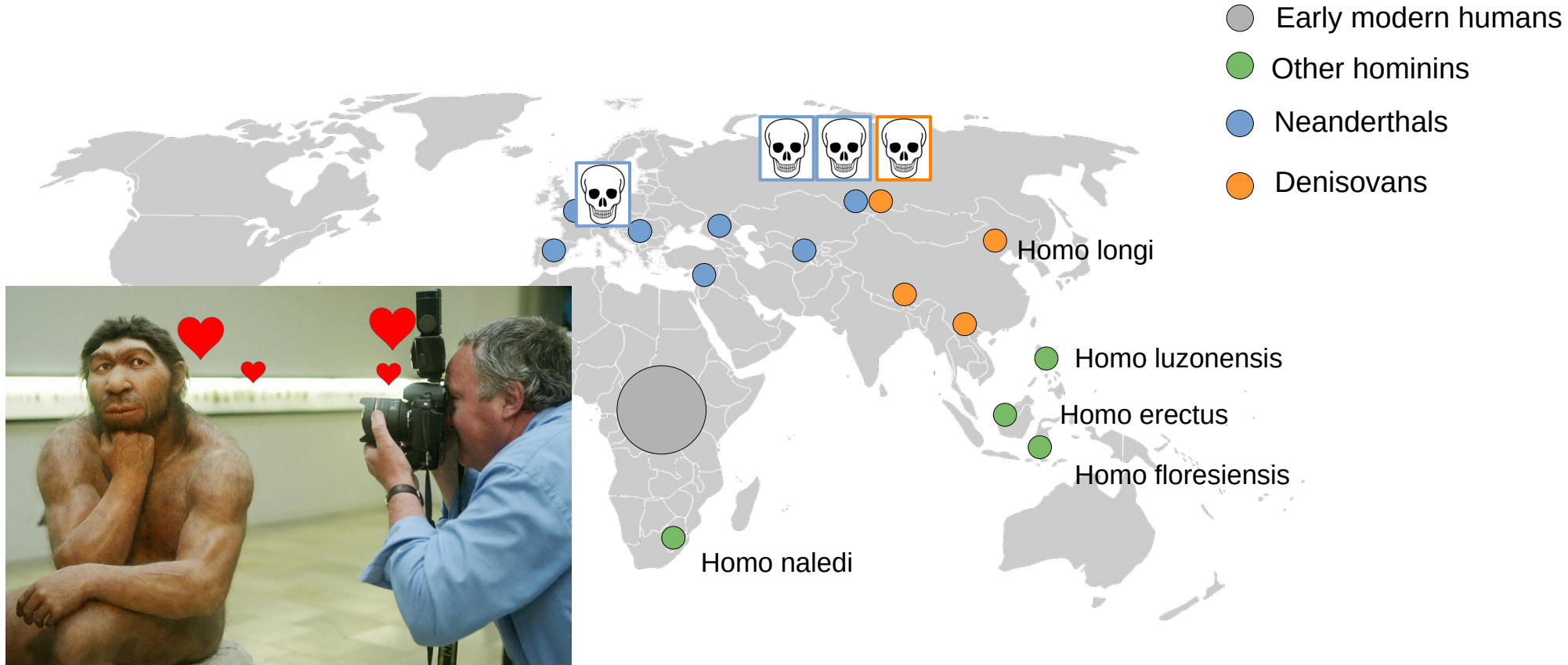
100 Thousand years ago there were at least 7 human groups alive



Now only homo sapiens remain



Our ancestors met and admixed with them before they went extinct



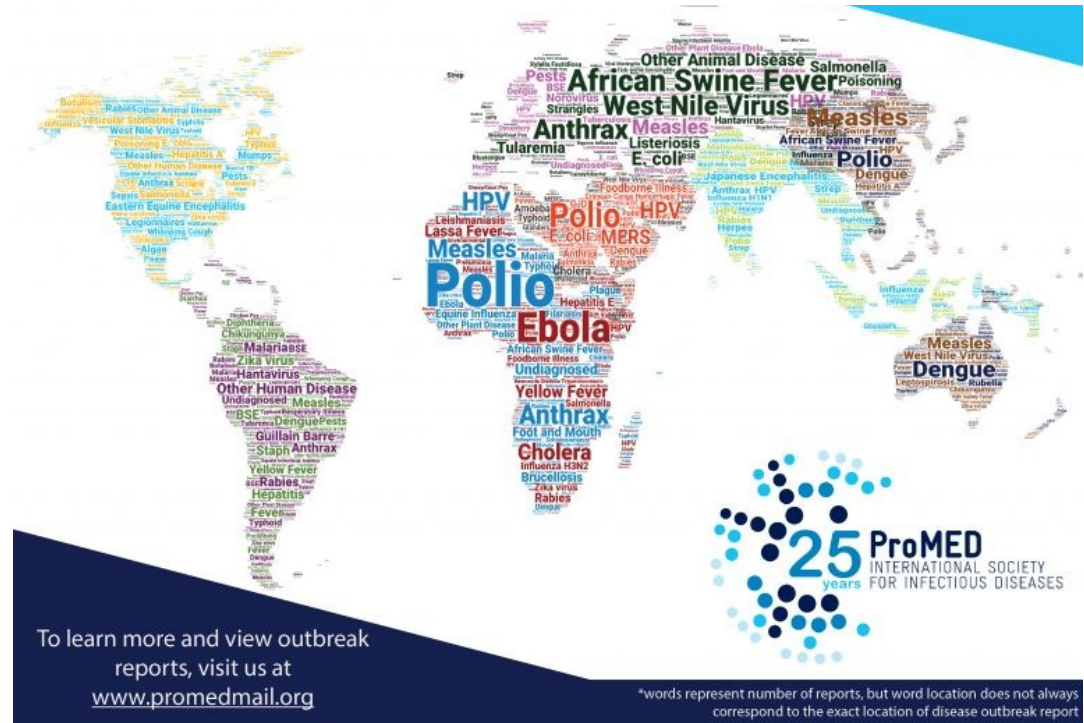
Adaptation to life in high altitudes

~86% frequency in people in Tibet



Huerta-Sánchez et al 2014

Lots of immune related genes in high frequency



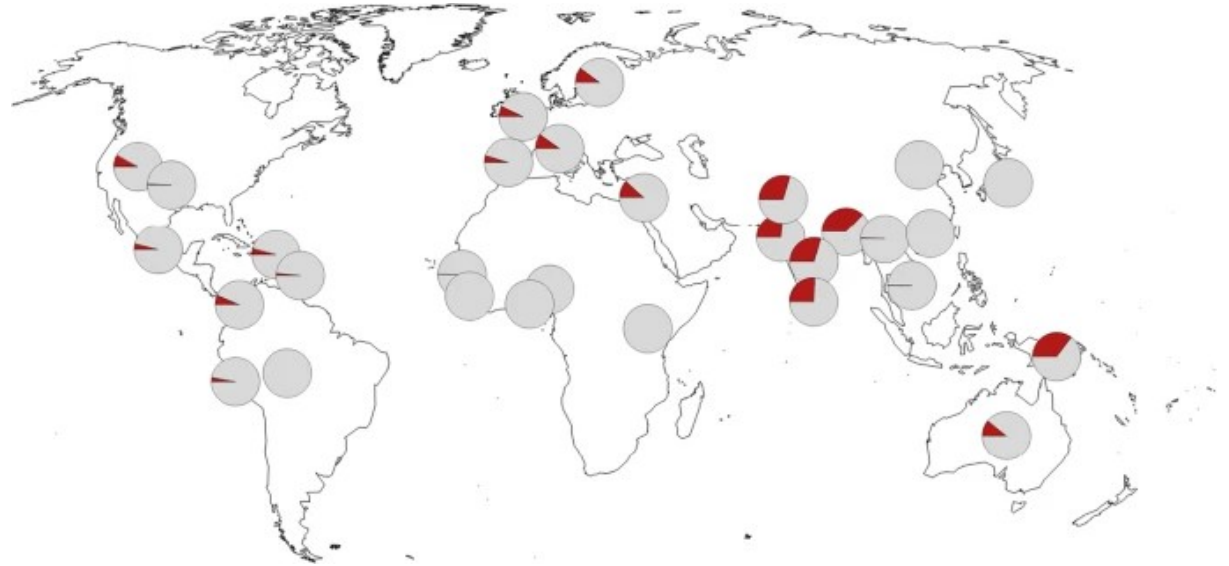
Dannemann and Kelso 2017
Sams et al 2016

Impacts of introgression

8% in Europe, 30% in South Asia



Increased risk of
hospitalization for Covid19



Zeberg and Paabo 2020

Archaic DNA lives on in us today

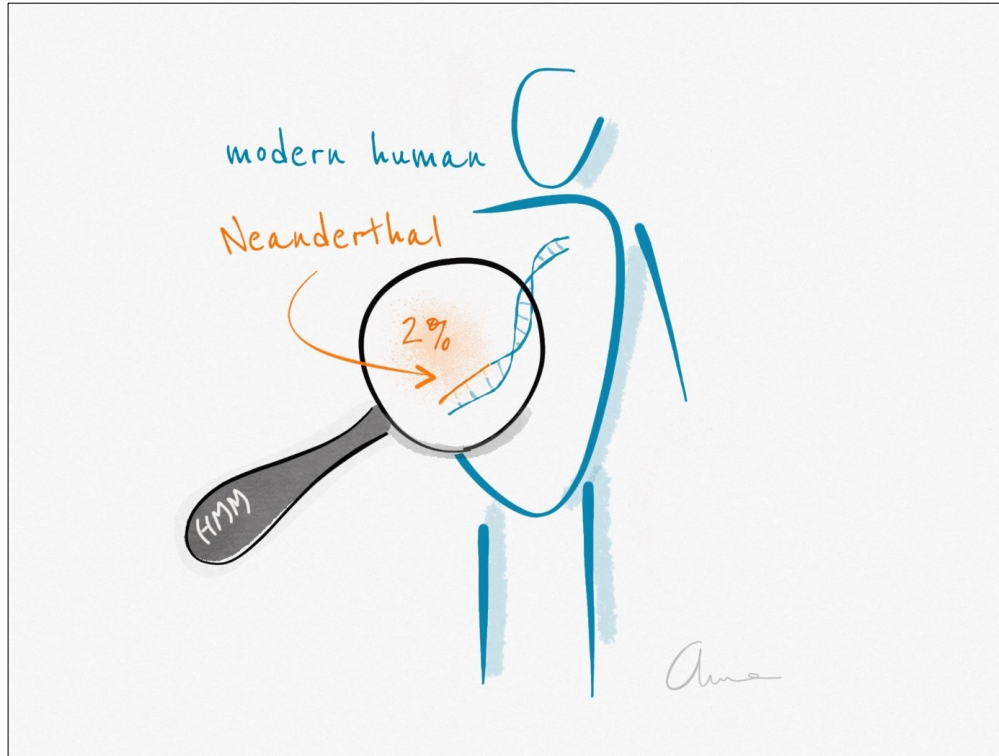


Which groups did we meet? Where did we meet them?



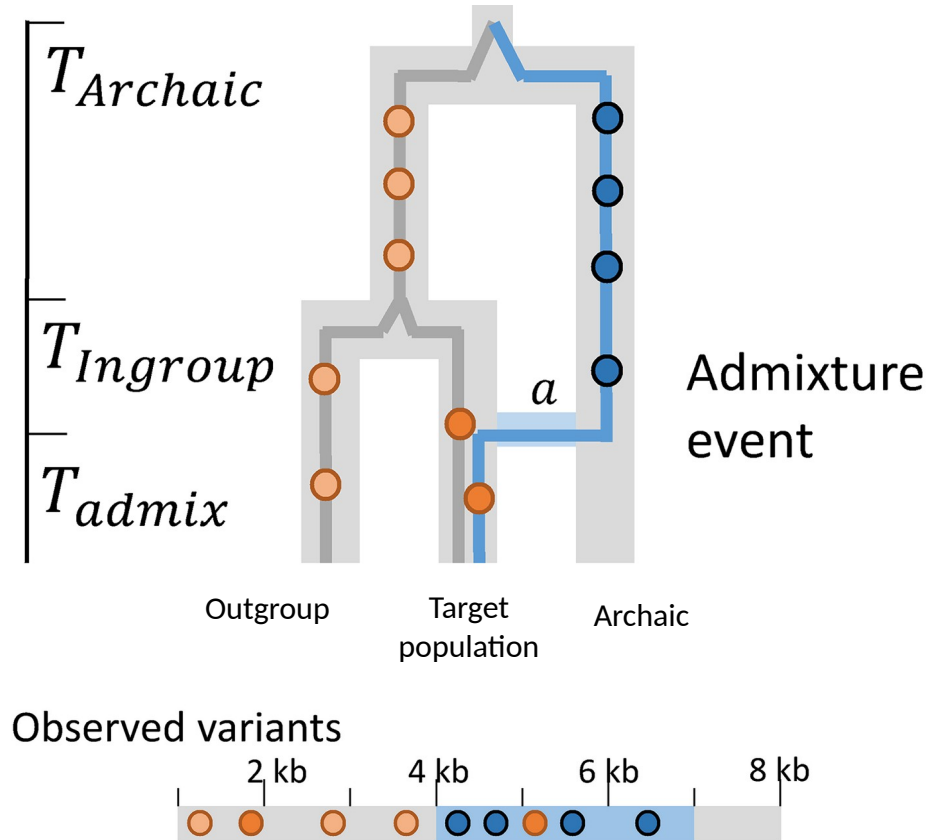
When did we meet them?

Finding archaic parts of our genomes

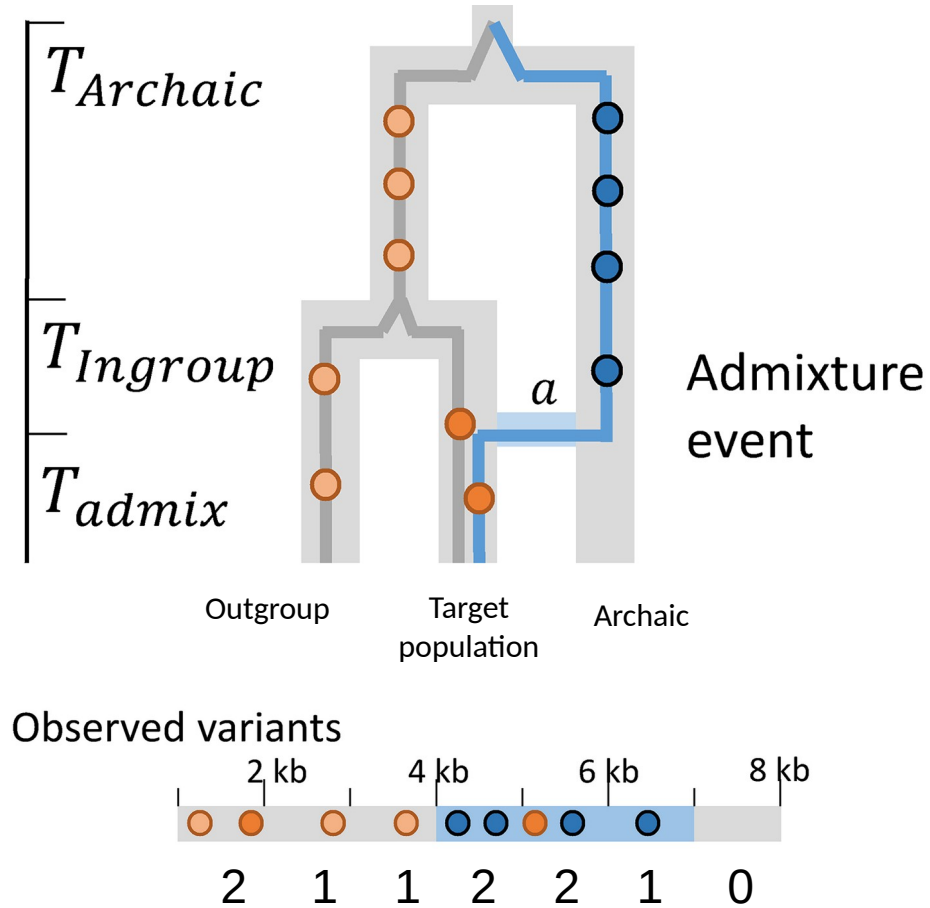


Identify archaic sequence in an individual's genome

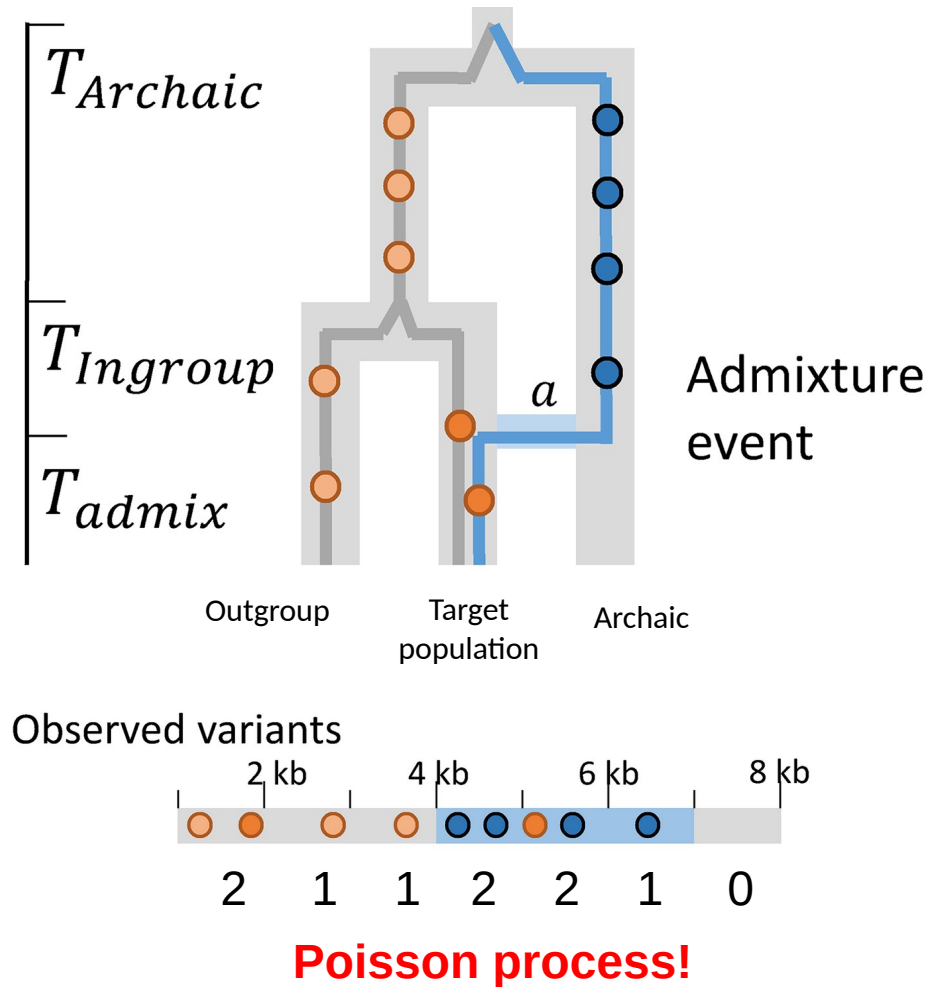
Finding archaic segments



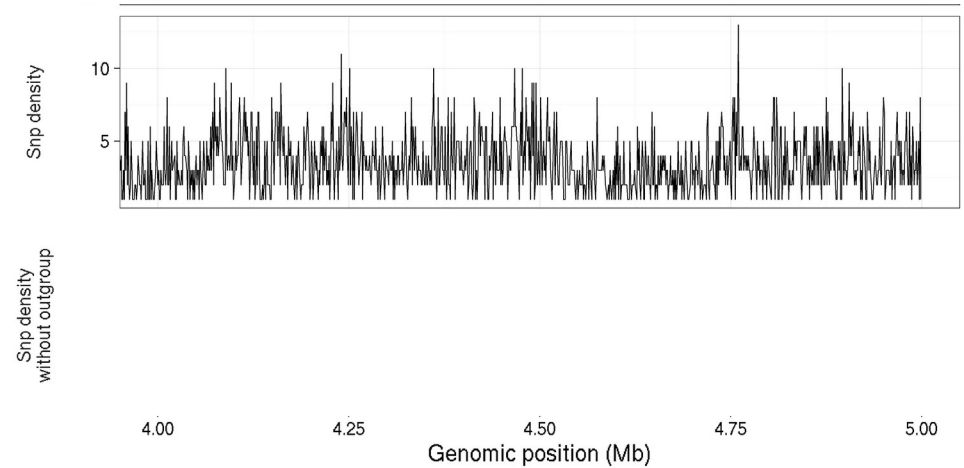
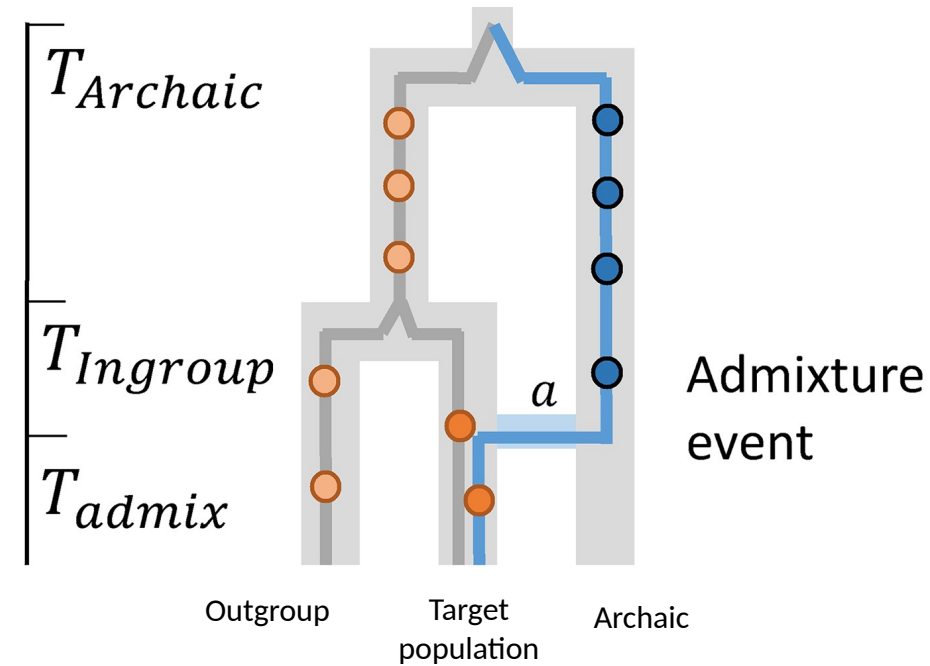
Finding archaic segments



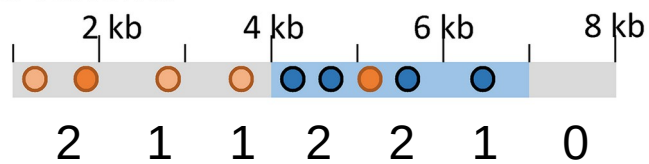
Finding archaic segments



Finding archaic segments

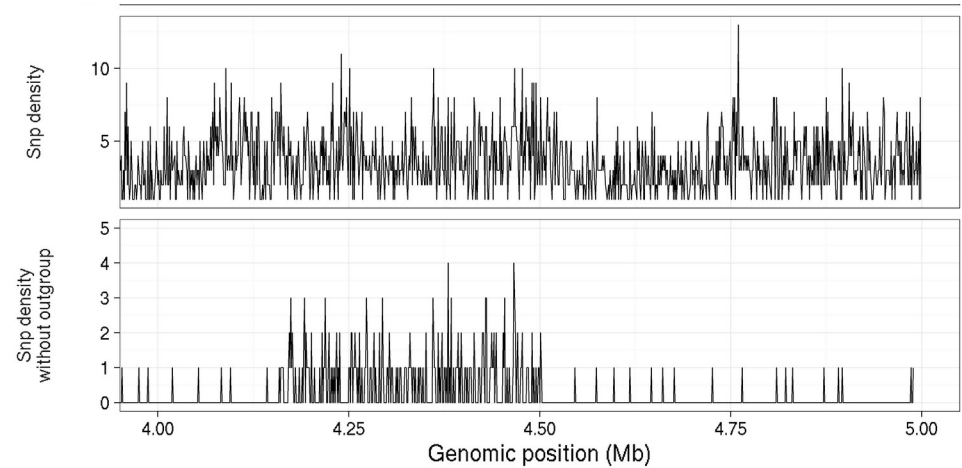
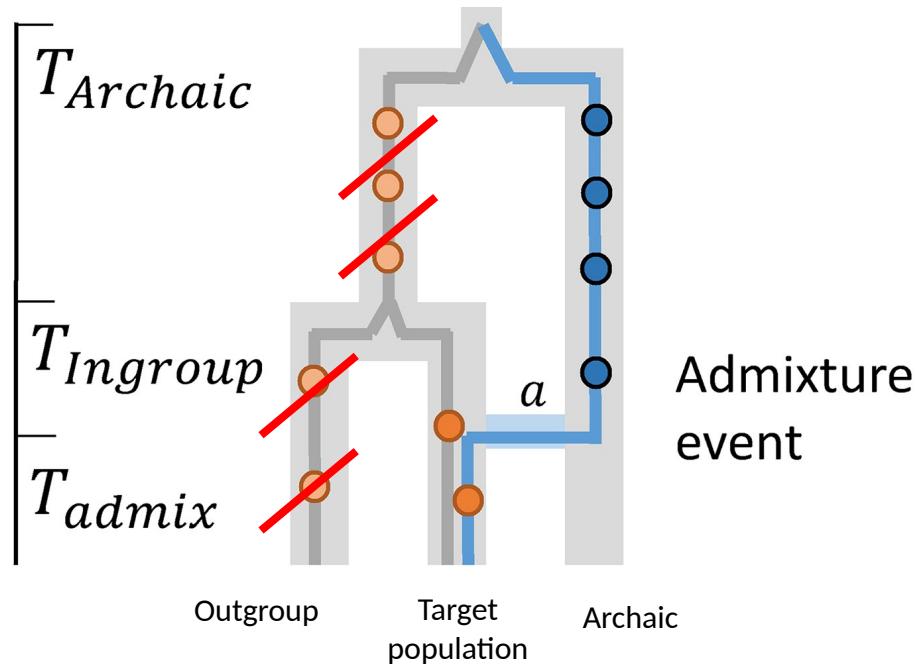


Observed variants

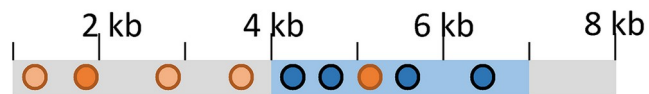


Poisson process!

Finding archaic segments



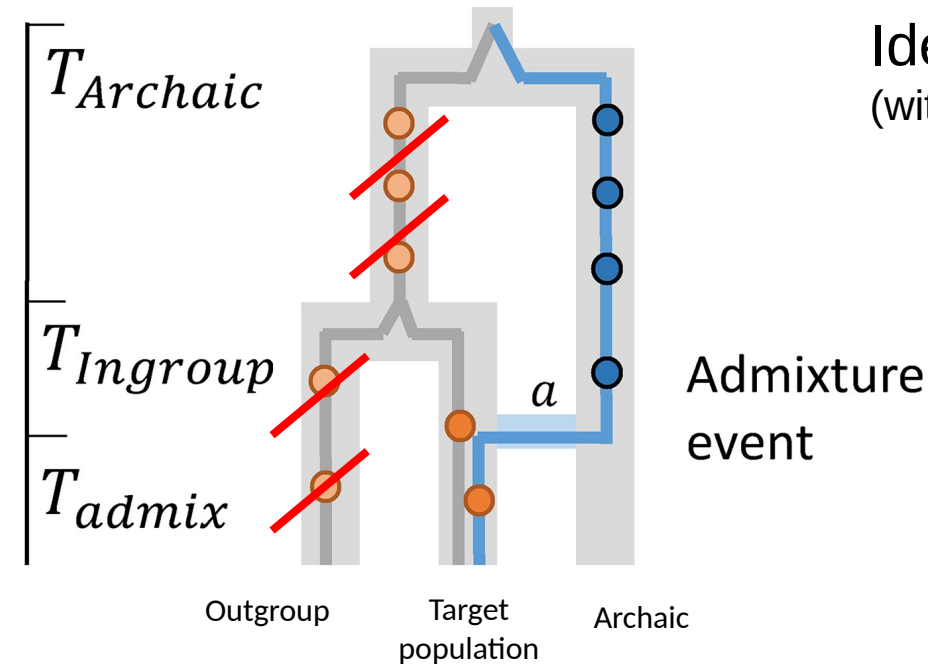
Observed variants



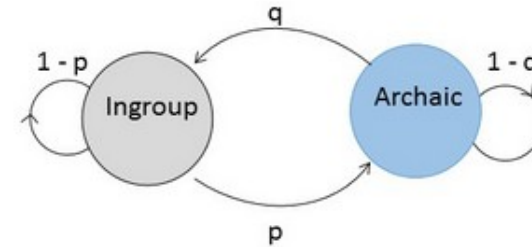
Remove variants found in outgroup



Finding archaic segments



Identify archaic segments with an HMM
(without using Neanderthal/Denisovan genomes)



Parameters

Emission parameters

$$\lambda_{Ingroup} = \mu \cdot L \cdot T_{Ingroup}$$

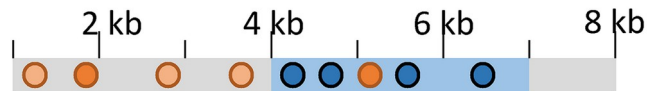
$$\lambda_{Archaic} = \mu \cdot L \cdot T_{Archaic}$$

Transition parameters

$$q = t_{admix} \cdot r \cdot L \cdot (1 - a)$$

$$p = t_{admix} \cdot r \cdot L \cdot a$$

Observed variants



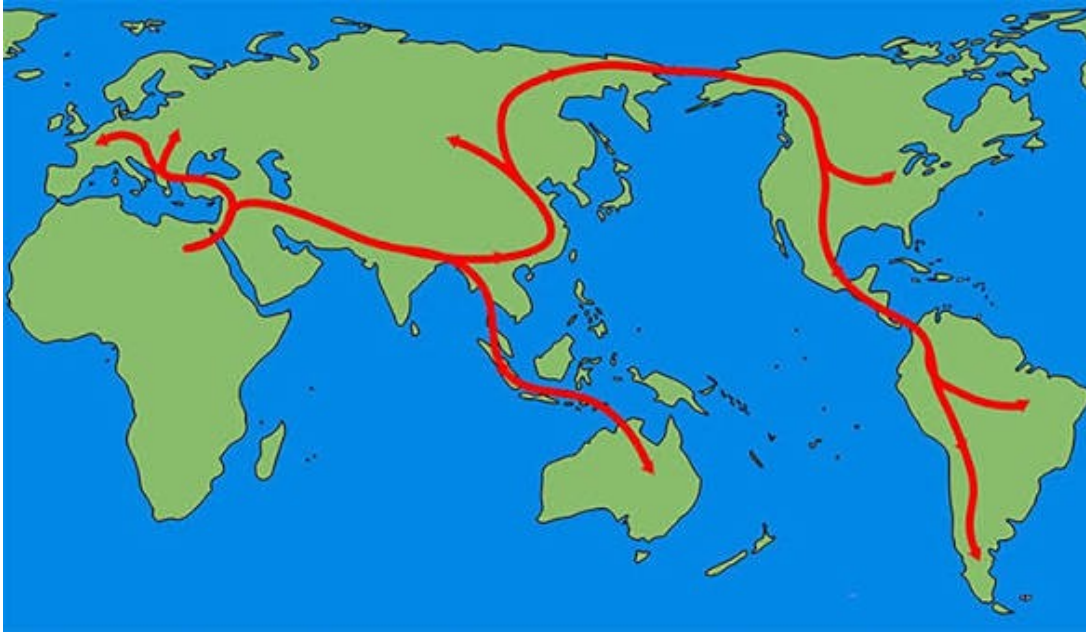
Remove variants found in outgroup



Emission (poisson means)

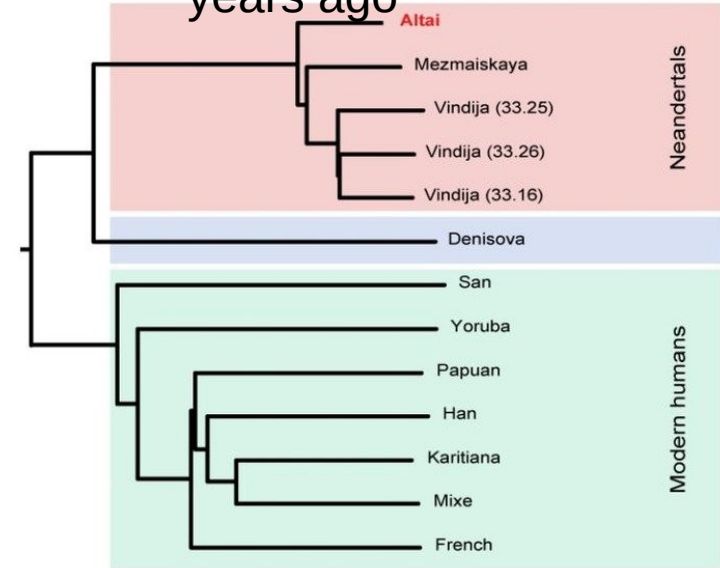
Ingroup/outgroup

Out of Africa event ~
50,000 years ago



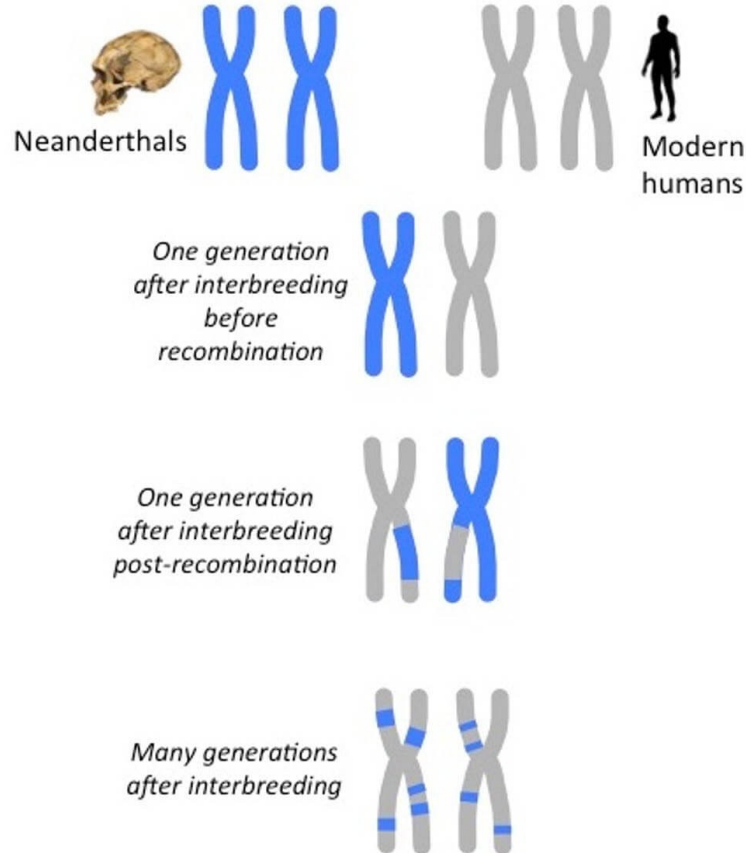
archaic/outgroup

Human/Neanderthal
ancestor ~ 680,000
years ago



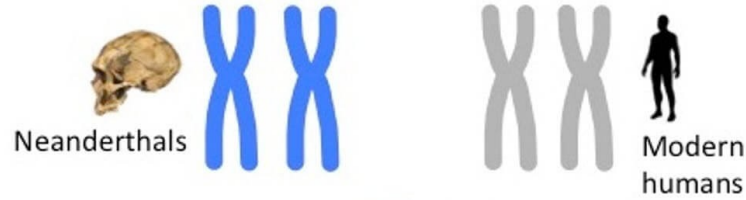
Transition parameters

Timing of neanderthal meeting



Transition parameters

Timing of neanderthal meeting



*One generation
after interbreeding
before
recombination*



*One generation
after interbreeding
post-recombination*



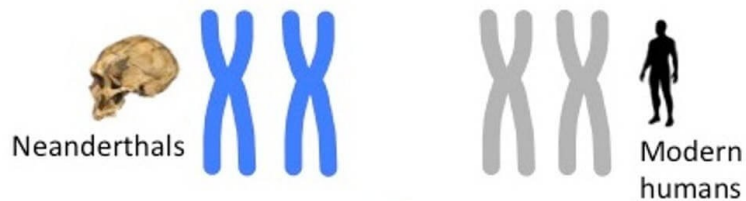
*Many generations
after interbreeding*



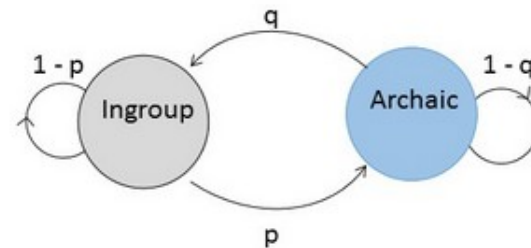
Geometric process!

Transition parameters

Timing of neanderthal meeting



Geometric process!



Parameters

Emission parameters

$$\lambda_{Ingroup} = \mu \cdot L \cdot T_{Ingroup}$$

$$\lambda_{Archaic} = \mu \cdot L \cdot T_{Archaic}$$

Transition parameters

$$q = t_{admix} \cdot r \cdot L \cdot (1 - a)$$

$$p = t_{admix} \cdot r \cdot L \cdot a$$

Humans and Neanderthals met around 47,000 years ago

